

120 Days of Quant Finance

Week 4 – Financial Theory

Day 1: Derivatives and Financial Instruments

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Introduction

Financial markets consist of a wide range of instruments with different payoff structures. In quantitative finance, instruments are best understood not as contracts or assets, but as *payoff functions* defined on future states of the world.

This section introduces primary financial instruments and derivatives, and presents a payoff-based framework that forms the foundation of no-arbitrage pricing and derivative valuation.

Classification of Financial Instruments

Financial instruments can be broadly classified into two categories:

- **Primary instruments**, whose payoffs arise directly from economic activity.
- **Derivative instruments**, whose payoffs depend on the value of other financial instruments.

This distinction is fundamental to financial theory and pricing.

Primary Financial Instruments

Equities

Equities represent ownership in a firm and entitle the holder to uncertain future cash flows. The terminal value of a stock is modeled as a random variable:

$$S_T \in \mathbb{R}_+.$$

Equity payoffs do not have a predetermined functional form and are determined by market forces.

Bonds

Bonds are fixed-income instruments with contractual cash flows. A bond typically pays coupons C_t at specified times and returns face value at maturity.

The total payoff can be written as:

$$\text{Bond Payoff} = \sum_t C_t + F,$$

where F denotes the face value.

Cash and Money Market Instruments

Cash and money market instruments are often idealized as risk-free assets. Their value evolves deterministically according to an interest rate r :

$$B_t = e^{rt}.$$

The risk-free asset serves as a reference unit (numéraire) in pricing theory.

Definition of a Derivative

A **derivative** is a financial instrument whose payoff depends on the value of one or more underlying variables, such as asset prices, interest rates, or indices.

Formally, a derivative payoff can be written as:

$$\text{Payoff} = f(S_T, K, T, \dots),$$

where $f(\cdot)$ is a deterministic function and S_T denotes the underlying price at maturity.

Derivatives themselves do not generate economic value but redistribute risk.

Economic Motivation for Derivatives

Derivatives are used for:

- Hedging unwanted risks,
- Transferring risk between market participants,
- Enhancing market completeness,
- Improving price discovery.

From a quantitative perspective, derivatives enable the construction of tailored payoff profiles.

Forward Contracts

A forward contract is an agreement to buy or sell an asset at a future time T for a predetermined price K .

The payoff of a long forward position at maturity is:

$$\text{Payoff}_{\text{forward}} = S_T - K.$$

Forward contracts have linear payoffs and, under no-arbitrage conditions, require no initial investment.

Futures Contracts

Futures contracts are standardized versions of forward contracts traded on exchanges. While pricing theory initially treats futures and forwards similarly, practical differences arise due to daily settlement and margining.

Options

Options provide the right, but not the obligation, to buy or sell an asset at a specified price.

Call Option

The payoff of a European call option with strike K is:

$$\text{Payoff}_{\text{call}} = \max(S_T - K, 0).$$

Put Option

The payoff of a European put option with strike K is:

$$\text{Payoff}_{\text{put}} = \max(K - S_T, 0).$$

Options exhibit nonlinear and convex payoff structures, distinguishing them from linear instruments.

Swaps

A swap is a contract in which two parties exchange streams of cash flows. Common examples include interest rate swaps, which exchange fixed-rate payments for floating-rate payments.

At a conceptual level, swaps can be viewed as portfolios of forward contracts.

Linear and Nonlinear Payoffs

Financial instruments can also be classified by the shape of their payoffs:

- **Linear payoffs:** stocks, bonds, forwards.
- **Nonlinear payoffs:** options and structured products.

Nonlinearity introduces complexity in pricing and hedging and motivates replication arguments.

Payoffs as Functions

In quantitative finance, every financial instrument is represented as a function of underlying variables:

$$\text{Instrument} \equiv f(S_T).$$

This functional perspective is essential for pricing, replication, and risk management.

Long and Short Positions

Every financial instrument has two sides:

- A **long position**, which receives the payoff.
- A **short position**, which pays the payoff.

Mathematically, the payoff of a short position is:

$$-f(S_T).$$

Consistency of pricing between long and short positions is enforced by no-arbitrage conditions.

Conclusion

Understanding financial instruments through their payoff structures is the foundation of derivative pricing theory. By viewing instruments as mathematical functions of future states, quantitative finance provides a unified framework for pricing, hedging, and risk management.

This payoff-centric perspective sets the stage for the no-arbitrage principle and risk-neutral valuation developed in subsequent sections.